Big Data and Machine Learning for Applied Economics

Week 09: The Final Project — Presentations and Course Closing

Eduard F. Martinez Gonzalez

Department of Economics, Universidad Icesi

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Today

8:00–8:15	The Problem Set 2 ranking and the course wrap-up in one slide
8:15–11:00	Project presentations: 20 minutes plus questions [individual: six of 20 + 5; in pairs: three of 30 + 15; TBD]
Today	Deliverable 2: the proposal document (8 pages max)

What you present today is not a report with results: it is a **predictive research proposal**, the design of a study that withstands the course's seven questions. If you have preliminary results, show them as what they are: preliminary.

The rule of the room

Every presentation ends with the room's questions before the instructor's. Everyone evaluates with the same list ("The questions we will ask you"); the grade is the instructor's.

Problem Set 2 debrief: the course ranking

Fifteen minutes [table and comments to be filled in after grading]

1. **The final standings:** each team's test RMSE for the course's final model, next to its validation RMSE and its Problem Set 1 one. Who moved up, who moved down, and why?
2. **Which model won:** boosting, the forest, an average of families, or the regularized model from the first problem set? How much flexibility was worth over the regularized regression.
3. **The comparison with the course's case:** with the polling stations, moving from the best linear model (census + department, 15.6) to boosting (12.8) was worth 2.8 RMSE points —flexibility—, and combining the two leagues left 11.4. Did housing behave the same? Explaining the difference is worth more than the number.
4. **The open box:** which blocks and variables carried the prediction in the best models, and whether the map rules the housing market the way it ruled the vote.
5. **The frequent stumbles:** a network without scaling or early stopping; SHAP on the training set; the test set opened twice.

1 The project presentations

2 Course closing

What the presentation must contain (40 project points)

The document's same seven sections, in 20 minutes:

1. **The question and the decision:** who would use the prediction and what they would do differently with it.
2. **The data:** source, unit, size, predictors by blocks; what would be known at prediction time.
3. **Metric, benchmark, and validation:** the metric the decision needs; the benchmark you would use without the model; the scheme that respects the data's structure.
4. **The methods and why:** at least two families from the course, and the reason for each.
5. **What would be interpreted, and what would not.**
6. **Risks:** leakage, extrapolation, regime change; fairness if people are involved.
7. **Work plan.**

How it is graded: coherence between the decision, the y, the metric, the benchmark, and the validation; methods justified with what the course covered; honesty about what the model will not say; and the answers to the questions. Slide design is not graded.

The questions we will ask you

The list the whole room evaluates with

1. What decision changes if the prediction is good? And if it is bad?
2. What information exists *at prediction time*? Which of your variables is leakage in disguise?
3. What is the benchmark? How much must the model beat it by to be worth it?
4. Does your validation respect the data's groups, time, or space? What question does it answer?
5. What would you do if the flexible model does not beat the simple one?
6. What will you interpret, with which tool, and which sentence do you refuse to write?
7. Where does the model degrade: which units, which period, which range of the x's?
8. If the project assigns something to people, which per-group metric will you report?

A proposal answering all eight is already a thesis chapter. Five is a good semester.

The document: the checklist before handing it in

- **8 pages max**, with the seven sections and their names; tables with units; legible figures.
- The y defined precisely (unit, timing, how it is measured) and the decision on page one.
- The **benchmark** made explicit, and the metric in the decision's units.
- The **validation scheme** drawn or described: what goes in each fold and why.
- The methods with their main hyperparameters and how they would be chosen.
- A **risks** paragraph with at least one concrete leakage you would avoid.
- The three sentences you will *not* write about the model (effects, individuals, forecasts).
- If there is preliminary data: split, recipe, benchmark, and test, with the problem sets' discipline.

Submission: the document is due today [time and channel to be confirmed]. It is worth 50 of the project's 100 points; the project is 60% of the course.

1 The project presentations

2 Course closing

What you take with you

A workflow, the same from session 2 through 7: split $\rightarrow$ preprocess inside the pipeline $\rightarrow$ validate according to the structure $\rightarrow$ tune on the training set $\rightarrow$ evaluate against the benchmark $\rightarrow$ interpret with caveats $\rightarrow$ report where it fails.

A kit of methods with their when: the line and the penalty when the signal is smooth or the data few; the logit and the cutoff when the y is a decision; trees and their ensembles on tabular data; PCA and k-means when there is no y; networks when the structure lives in the representation, or can be borrowed.

An attitude:

- Y, not β: predicting well and explaining well are different questions.
- The table decides, on the same test set, with intervals, and only once.
- Significant is not predictive; important is not causal; the Lasso selects predictors, not causes.
- A CV number answers its scheme's question.
- To represent is to compress with purpose.

What we did not cover, and where to go next

- **Text and language models:** counting words turns a document into a row with thousands of columns, and from there the whole course applies; embeddings and LLMs are borrowed representations validated against human labels. Gentzkow, Kelly & Taddy (2019); Dell (2025).
- **Causal machine learning:** double selection, double machine learning, and causal forests: ML predicts the nuisances and the economist supplies the identification. Athey & Imbens (2019); Chernozhukov et al. (2018). The Causal Inference course is the place.
- **Forecasting and horse races:** many families on long series, with forward validation. Medeiros et al. (2021); Gu, Kelly & Xiu (2020); Goulet Coulombe et al. (2022).
- **Fairness and algorithmic decisions:** which per-group metric, and why you cannot have them all. Kleinberg, Mullainathan & Raghavan (2017); Obermeyer et al. (2019).
- **What stayed in the appendices:** nested CV, Bayesian search, BART, SVM, Gaussian mixtures, Adam. They are in the decks for when you need them.

Closing

- **Grades:** the Problem Set 2 ranking closes today; the project grade comes from the document (50), the presentation (40), and the November 3 pitch (10). Final grades in the system from December 1.
- **The materials** stay in the course repository: the decks, the R applications with their data, and the assignments. The session scripts are the starting point for any project you pursue.
- **The course evaluation** in the institutional system: read it twice, like Molnar's disadvantages.
- **Thank you.** A prediction course ends the way it should: with the test set, used once.

Questions and presentations. . .

Essential course references

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